RESUME

DR. MAMTA AGRAWAL **Email:** mamta.pankaj23@gmail.com
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Career Objective

To leverage my extensive teaching and research experience in Applied Mathematics and Computational Biology to contribute significantly to academic excellence and innovation. As an associate professor and researcher with expertise in mathematical and computational modeling, I aim to advance interdisciplinary research, mentor future scholars, and foster impactful collaborations globally. With a strong background in project management, international exposure, and contributions to reputed journals and conferences, my objective is to continue addressing complex problems in mathematical sciences while contributing to the growth and reputation of my institution.

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BHOPAL, M.P.-
462016, INDIA

Personal Data:

DOB :23-11-1969
Sex: Female
Languages: English and Hindi.
Nationality: Indian
Marital Status: Married
Hobbies: Painting, Music & Doing Exercise and Meditation
Strengths: Optimistic Perseverance, Team Leader cum Team player

Work Experience – More than 20 years

(2008 to 2016 worked as a PHD fellow researcher and Women scientist of WOS-A DST New Delhi.)

1. Working as an Associate Professor, In the Department of Mathematics, School of Applied sciences and language, Vellore Institute of Technology, Bhopal University, From 12 June 2017 to till date.

2. Worked as a Women Scientist Principal Investigator (PI) under Women Scientist Scheme (WOS-A) of DST, New Delhi, in the Department of Mathematics Maulana Azad National Institute of Technology, MANIT, Bhopal from 31st May 2013 to 31st May 2016.

   Principal Investigator (PI) of the project Dr. Mamta Agrawal

   Mentor of this project: Dr. KR Pardasani.

3. Worked as a Contract lecturer of Mathematics in the Department of Mathematics, MANIT, Bhopal from 16th January 2012 to 31st May 2013.

4. Joined PhD as a regular research fellow on 1st September 2008 in the Department of Mathematics, MANIT, Bhopal and submitted PhD thesis on 30th August 2011.

5. Worked as a Contract lecturer of Mathematics in the Department of Mathematics, MANIT, Bhopal from 10th July 2007 to 31st August 2008.

6. Worked in many colleges from 2003 to July 2007.

Educational Details

Post Doctorate (Computational and Mathematical Biology)
University : MANIT, Bhopal (Under WOS-A)
Year : 2013-2016

Ph.D (Mathematical Sciences)
University : MANIT, Bhopal,
Percentage : 9.38 CGPA
Awarded : 2012

MCA (Computer Application)
University : BU, Bhopal
Result : 1st Division
Passing Year : 1998

PGDCA (Computer Application)
University : BU, Bhopal
Result : 1st Division
Passing Year : 1994

M.SC (Mathematics)
University : BU, Bhopal
Result : 1st Division
Passing Year : 1991

B.SC (Maths, Physics, Chemistry)
University : Barakatullha University, BU, Bhopal
Result : 1st Division
Passing Year : 1989

Responsibility Handled :-

Currently handling in VIT	
1.	. Guiding 04 Ph.D Research Scholar, Project development guidance for bachelors & masters students.
2.	Research activity, handling class, taking lecture and assessment activity for students.
3.	Serving as the Head of the Fine Arts Club.
4.	Serving as the Head of PhD panel.
Handled	
1.	Developed syllabus of heat and mass transfer course in VIT Bhopal.
2.	Fine arts club coordinator from 2017 to till now.
3.	Member in BOS in VIT Bhopal in 2018-2019.
4.	Phd selection committee member.

PUBLICATIONS DETAILS:-

PAPERS PRESENTED IN CONFERENCES: 40
CONFERENCES - WORKSHOP ATTENDED: 47
RESEARCH PAPERS PUBLISHED: 24

Chapter Published- 01

Applications of birth and death process in biology, Research trends in Mathematics and Statistics,Volume-9, Ankit Publications, Volume -9, ISSN NO:978-93-90541-03-4, Publications year:2020

Book Published – 01

1	**Book Title** - Some useful Transforms and Partial Differential Equations (For B.Tech/B.E Students & Other Competitive Exams) **Authors:** A.Manickam, Mamta Agrawal, **Coauthor:** Shweta Singh **ISBN - :**978-81-950760-6-2 **DOI -** https://doi.org/10.59646/ca/242 **Link - https://doi.org/10.22271/ed.book.999**

Patent Published-

1	**Title** - "A LIGHTENING ALERT SYSTEM FOR FARMERS" **Patent Published –** 30 September 2022

Research Papers Published - 30

S.No.	Authors	Year	Title of Paper	Journal	Volume	Pages
1.	Mamta Agarwal, A.Manickam, Ajinkya Deokate S.Gharote, A. Ojha and P. Nagrale	2024	Investigatory analysis of the natural hazards on the Indian coastline	Mathematical modeling and Computing	DOI:10.23939/mmc2024.03.720 http://www.scopus.com/inward/record.url?eid=2-s2.0-85205719681&partnerID=MN8TOARS	720-729
2.	Venkanna Bachu and Mamta Agrawal	2024	Approximation of temperature distribution in the tissue of the human body by finite difference method	Revista investigacion operacional	Volume: 45 http://www.scopus.com/inward/record.url?eid=2-s2.0-85186073264&partnerID=MN8TOARS	8-13
3.	Mimansha Agrawal, Ajinkya Deokate, Mamta Agrawal, Zafer Aslan	2024	Impact of Wind and Temperature on Covid-19 Outbreak: Investigatory Analysis ın India	EURAS Journal of Engineering and Applied Sciences	Volume: 4 Issue: 1 http://dx.doi.org/10.17932/ejeas.2021.024/ejeas_v04i1001	1 - 15

4.	Mamta Agrawal	2022	Effect of Climate on Human Health	IGCC 2022 Conference Publication	Web: https://igcc2022.aydin.edu.tr/	30 - 35
5.	Mimansha Agrawal, Mamta Agrawal, Zafer Aslan, İlknur DÖNMEZ, Ali GÜNEŞ	2022	Monitoring Air Pollution Impacts of COVID- 19 in India	Euras Journal Of Health	Volume: 2 Issue: 2 December - 2021 http://dx.doi.org/10.17932/ejoh.2020.022/ejoh_v02i2001	93- 103
6.	Mimansha Agrawal, Ajinkya Ravindra Deokate, Mamta Agrawal	2021	Atmospheric Factors On Land Surface Temperature	OSTIV (Organisation Scientifique et Technique Internationale du Vol à Voile), Turkey	Volume: 45 Issue: 2	3
7.	A. Leema Rose , A.Manickam, Mamta Agrawal	2021	A Mathematical Model For The Special Effects Of Phosphatidylserine On Endocrine Reaction To Reasonable Concentration Exercise In Healthy Male Subjects	Turkish Journal of Computer and Mathematics Education	ISSN: 1309-4653 Volume: 12 Issue: 3 http://dx.doi.org/10.17762/turcomat.v12i3.1632	3555-3559
8.	R. Kalaiselvi, A. Manickam and Mamta Agrawal and pushpendra kumar	2021	A study of mathematical model for Extended Lognormal Distribution to obligatory Role of Hypothalamic Ovariectomized Rhysus Monkey	Annals of the Romanian society for cell Biology	ISSN: 1583-6258 Volume : 25 Issue: 5	4122-4127
9.	Kalaiselvi .R, Manickam ,A and Mamta Agrawal	2021	A mathematical model for maximum Likelihood Estimation to Atrazine Inhibits pulsatile Gonatropin- Releasing Hormone in Animals	Turkish Journal of Computer and Mathematics Education	ISSN: 1309-4653 Volume : 12 Issue: 3 https://api.semanticscholar.org/CorpusID:237249008	3550-3554
10.	R. Kalaiselvi, A. Manickam and Mamta Agrawal	2021	A mathematical model for pulsatile gonadotropin releasing hormone release from hypothalamic explants of male marmoset monkeys compare with male rats	Advances in Mathematics: Scientific Journal 10 "Scopus"	Volume 10 No.1 ISSN:185 7-8365 http://www.scopus.com/inward/record.url?eid=2-s2.0-85100827616&partnerID=MN8TOARS	629-636
11.	A Leema Rose, A. Manickam and Mamta Agrawal	2021	A mathematical model For stimulated hypothalamic pituitary adrenal reactivity dynamic changes in healthy patients	Advances in Mathematics: Scientific Journal 10 "Scopus"	Volume 10 No.1 ISSN: 1857-8365 http://www.scopus.com/inward/record.url?eid=2-s2.0-85100594622&partnerID=MN8TOARS	Pages 637-644

12.	A. Manickam and Mamta Agrawal	2020	Research Trends in Mathematics and Statistics. Chapter - 7 Applications of Birth and Death Process in Biology	Published By: AkiNik Publications Total Pages: 153 ISBN: 978-93-	Volume – 9 Page 83 ISBN: 978-93- 90541- 03-4 https://doi.org/10.22271/ed.b ook.999	Total Pages: 153
13.	Mamta Agrawal, KR Pardasani	2019	Finite Element Model to Study the Effect of Lipoma and Liposarcoma on Heat Flow in Tissue Layers of Human Limbs	Lecture notes in Mechanical Engineering (**Springer**) "Numerical Heat Transfer and Fluid Flow". "Proceedings of NHTFF 2018"	eBook ISBN 978- 981-13- 1903-7 http://www.scopus.com/inward/record.url?eid=2-s2.0-85058948478&partnerID=MN8TOARS	241- 247
14.	Mamta Agrawal, KR Pardasani	Nov. 2018	Numerical Model To Study Thermal Changes Due To Sarcoma In Tissues Of Human Limbs	"Advances and Applications in Mathematical Sciences"	Volume:18, Issue 1, https://www.mililink.com/upload/article/412524118aams_vol_181_nov_2018_a13_p169-180_mamta_agrawal_and_k.r._pardasani.pdf	169- 180
15.	Mamta Agrawal, KR Pardasani	2016	Finite element model to study temperature distribution in skin and deep tissues of human	Journal of Thermal Biology **Elsevier SCI**	Volume:62 http://www.scopus.com/inward/record.url?eid=2-s2.0-84994074772&partnerID=MN8TOARS	98-105
16.	Mamta Agrawal, Neeru Adlakha, KR Pardasani	2015	Finite element model to study the thermal effect of tumors in dermal regions of irregular tapered shaped human limbs	Journal of Thermal Sciences **Elsevier SCI**	Volume: 98 http://www.scopus.com/inward/record.url?eid=2-s2.0-84939448136&partnerID=MN8TOARS	287- 295
17.	Mamta Agrawal, Neeru Adlakha, KR Pardasani	2014	Steady state temperature distribution in dermal regions of an irregular tapered shaped human limb with variable eccentricity	Journal of Thermal Biology **Elsevier SCI**	Volume: 44 http://www.scopus.com/inward/record.url?eid=2-s2.0-84903889428&partnerID=MN8TOARS	27-34
18.	Mamta Agrawal et al.	2013	On Some Probabilistic Aspects of Diffusion Models for Tissue Growth	The Open Statistics and Probability Journal	Volume: 5 http://dx.doi.org/10.2174/1876527001305010014	14-21
19.	Mamta Agrawal, Neeru Adlakha, KR Pardasani	2011	Finite Element Model to Study Thermal Effect of Uniformly Perfused Tumor in Dermal Layers of Elliptical Shaped	International Journal of Biomathematics **SCI World Scientific**	Volume: 4 http://www.scopus.com/inward/record.url?eid=2-s2.0-84857602207&partner	1–14

			Human Limb		ID=MN8TOARS	
20.	Mamta Agrawal, Neeru Adlakha, KR Pardasani	2011	Effect of Tumor on Thermal Distribution in Elliptical Shaped Human Limbs	Proceeding of IC2It 2011. The 7th Int. Con. On Computing and Information Technolgoy IEEE Thailand.	Conference Paper Proceeding	213-217
21.	Mamta Agrawal, Neeru Adlakha, KR Pardasani	2011	Modelling and Simulation on Thermal Effect in Human Limbs	2011 Int. Symposium on Devices Intelligent system and Communication ISBN No. 978-93-807480-2 **MEMS, Intelligent Systems & C ommunication 2011 by IJCA Journal Number 8-Article 1 Year of Public ation: 2011**	**ISSN : No 0975-8887**	
22.	Mamta Agrawal, Neeru Adlakha, KR Pardasani	2010	Three Dimensional Finite Element Model to Study Heat Flow in Dermal Regions of Elliptical and Tapered Shape Human Limbs	Applied Mathematics and Computation **Elsevier SCI**	Volume: 217 http://www.scopus.com/inward/record.url?eid=2-s2.0-78649962098&partnerID=MN8TOARS	4129–4140
23.	Mamta Agrawal, Neeru Adlakha, KR Pardasani	2010	Seminumerical Model to Study Temperature Distribution in Peripheral Layers of Elliptical and Tapered Shaped Human Limbs	Journal of Mechanics Medicine and Biology **SCI World Scientific**	Volume: 10 http://www.scopus.com/inward/record.url?eid=2-s2.0-77951807241&partnerID=MN8TOARS	1–163
24.	Mamta Agrawal, Neeru Adlakha, KR Pardasani	2010	Thermal Disturbances in Dermal Regions of Human Limbs Involving Metastasis of Tumors	International Mathematical Forum	Volume: 39 https://www.m-hikari.com/imf-2010/37-40-2010/agrawalIMF37-40-2010.pdf	1903 – 1914
25.	Mamta Agrawal, Neeru Adlakha, KR Pardasani	2010	A Mathematical Model to Study Temperature Distribution in Elliptical Shaped Human Limbs Involving Non uniformly Perfused Tumor	Proceedings Of International Conference on Mathematical Computing and Management ICMCM	Confrence Paper ICMCM	S.No. 20
26.	Mamta Agrawal, Neeru	2009	Three Dimensional Seminumerical Model to	Applied Mathematical	Volume: 3	2625 –

	Adlakha, KR Pardasani		Study Heat Flow in Peripheral Layers of Elliptical Shaped Human Limbs	Sciences **SCImago**	http://www.scopus.com/inward/record.url?eid=2-s2.0-77950949731&partnerID=MN8TOARS	2638
27.	Mamta Agrawal, Neeru Adlakha, KR Pardasani	2009	Three Dimensional Finite Elements Model to Study Temperature Distribution in Peripheral Regions of Elliptical Shaped Human Limbs.	International Journal of Applied Science and computations **American Mathematical Society**	Volume: 16	91 – 105
28.	Mamta Agrawal, Neeru Adlakha, KR Pardasani	2009	Two Dimensional Thermal Distribution Model in Dermal Layers of Elliptical Shaped Human Limbs Involving a Uniformly Perfused Tumor	International Journal of Computational and Applied Mathematics	Volume: 4 https://www.researchgate.net/publication/228684108_Two_Dimensional_Thermal_Distribution_Model_in_Dermal_Layers_of_Elliptical_Shaped_Human_Limbs_Involving_a_Uniformly_Perfused_Tumor	**37–48**
29.	Mamta Agrawal, Neeru Adlakha, KR Pardasani	2009	Finite element model of thermograms for elliptical shaped human limbs involving tumor	Proceedings of All India Seminar on Life Seminar on Life Cycle Analysis, Measurement and Condition Monitoring	Confrence Paper Proceeding	100-109
30.	Mamta Agrawal, Neeru Adlakha, KR Pardasani	2008	Cubic Splines and Fourier Series Approach to Study Temperature variation in Dermal Layers of Elliptical Shaped Human Limbs	International Journal of Computational and mathematical Sciences SCImago	Volume: 2 https://api.semanticscholar.org/CorpusID:7585076	206-210

List of Projects Completed and Grants :-

1. Indo-Austria (DST–BMWF) Joint Research project sponsored by DST, New Delhi, Duration of the project: 2 Years+ extended (31st March 2011 to December, 2013).
   Total Cost of the Project: Rs. 6,38,400 /-
   Title of the project: "Thermal Modelling of Cancer"
2. Women Scientists Scheme A (WOS-A) research project by Government of India Ministry of Science & Technology (DST), New Delhi.

Duration of the project : 3 Years (31st May 2013 to 31st May 2016) Total Cost of the Project: Rs. 21,09,355 /-

Title of the Project: "Computational Models to Study Temperature Distribution in Invivo tissues of extended cylindrical human organs involving malignant tumors."

Grants :-

1. DST Grants for Organizing International Conference ICGAMS 2K19 in VIT Bhopal Rs- 1,50,000/-
2. TIMC Grants for Organizing International Workshop ICGAMS 2K19 in VIT Bhopal Rs- 25,000/-

Papers Presented at Conferences and Symposium:-

S. No.	Title of the Paper	Authors	Name of the Conference and Year
1.	Atmospheric Factors on Land Surface Temperature	Mamta Agrawal, Mimansha Agrawal, Ajinkya Ravindra	OSTIV MET PANEL 2021 17-18 February 2021 Istanbul Aydın University, Istanbul, TURKEY
2.	Analytical Approach to Study the effects of Physical and Physiological parameters on Temperature Distribution in Human organs	Mamta Agrawal, Venkanna bachu	Andhra Loyola College Vijayawada International e-conference on Advances in Mathematical SciencesJanuary 8-9 2021
3.	Analytical Study of Role of Evaporation, Physical and Physiological parameters on Temperature Distribution in Human Dermal Region	Mamta Agrawal, Venkanna bachu	PSG College of Technology,Coimbatore, India International conference on Applied Mathematical Models (ICAMM 2021)-Online January 7-9 2021
4.	The Mathematical is Model to Study the Effect of Clothes on Counter Current Heat and Water Distribution in Human Organs	Mamta Agrawal, Varsha Nigam,	VIT Bhopal University, 2nd National Conference RTASCE-2020, 18-20 Dec 2020
5.	Effect of Mask on Temperature Flow in Human Organs	Mamta Agrawal, Ajinkya Ravindra Deokate,	VIT Bhopal University, 2nd National Conference RTASCE-2020, 18-20 Dec 2020
6.	A Numerical Study of Heat Flow in Female Human Organs Body Before and After Menstrual Cycle	Mamta Agrawal, Laxmi	VIT Bhopal University, 2nd National Conference RTASCE-2020, 18-20 Dec 2020
7.	Mathematical Model to Study the Effects of Various Physical and Psychological Parameters on Temperature Distribution in Human Body	Mamta Agrawal, Venkanna bachu	VIT Bhopal University, 2nd National Conference RTASCE-2020, 18-20 Dec 2020
8.	Mathematical Model to study Heat and Water Distribution in Human Dermal Regions of Upper Limb	Mamta Agrawal, Venkanna bachu	VIT Bhopal University, International Conference of Gwalior Academy of Mathematical Sciences ICGAMS-2K19, 13-15 Dec 2019
9.	A Numerical Model to Study Thermal Disturbances Due to Presence of Sarcoma in Human Limbs	Mamta Agrawal	Anand Vihar College for Women, Bhopal National Seminar Value Based Education and its Role in Quality Assurance of Higher Education Institutes 18th April 2019
10.	Mathematical Model to study deep tissue and surface temperature Profiles involving normal and abnormal region of the limb	Mamta Agrawal	ICIREMPS 2K18-19 Sagar Institute of Research and Techonology, SIRT, Bhopal, 21-23th Feb 2019
11.	A numerical model to study heat and water distribution in human dermal regions of upper limb.	Mamta Agrawal	TIT Bhopal International Conference on Emerging Trends in Computing Technology and Management 7-9 Feb 2019.
12.	Numerical Model To Study Thermal	Mamta	International Conference on

	Changes Due To Sarcoma In Tissues Of Human Limbs	Agrawal, K.R. Pardasani	Frontiers in Industrial and Applied Mathematics FIAM-2018 April 26-27, 2018, NIT Hamirpur (H.P.)
13.	Finite Element Model to Study the Effect of Lipoma and Liposarcoma on Heat Flow in Tissue Layers of Human Limbs	Mamta Agrawal, K.R.Pardasani	International Conference Numerical Heat Transfer and Fluid Flow, NIT, Warangal, India January 19-21, 2018
14.	Computational Model to Study Temperature Distribution In Deep Tissue of Human Limb Containing Liposarcoma Tumor	Mamta Agrawal	International Symposium on operations research and game theory: modeling and Computation (ISORGT18), Indian Statistical Institute (ISI) Delhi Centre, January 9-11, 2018
15.	A computational model to study temperature distribution in dermal regions of human limbs involving tumors	Mamta Agrawal, K.R. Pardasani	IWM Regional Workshop on Research and Opportunities at IISER Bhopal during October 7-8, 2017.
16.	A mathematical model to study temperature distribution in deep tissues of elliptical shaped human limbs involving tumor	Mamta Agrawal, K.R. Pardasani	International Conference on Applications of Mathematics to Nonlinear Sciences (AMNS-2016) May 26-29, 2016, Kathmandu, Nepal
17.	A Steady State Finite Element Model To Study Temperature Distribution In Deep Tissues Of Elliptical Shaped Human Limbs	Mamta Agrawal, K.R. Pardasani	International Conference on Advances in Electronics, Computer and Mathematical Sciences ICIREMPS 2k16 organized by Sagar Society of Interdisciplinary, Research and Technology at Sagar Institutions of Research, Technology and Science, Bhopal during February 27-28th , 2016.
18.	Two dimensional finite element model to study temperature distribution in deep tissues of human limbs	Mamta Agrawal, K.R. Pardasani	18th annual cim 1st conference of Vijhnana Parishad of India on Computaional and interactive sciences and International symposium on computational Biology. 11-14 December 2015
19.	Finite element model to study temperature distribution in dermal regions of irregular tapered cylindrical shaped human limbs.	Mamta Agrawal, K.R. Pardasani	International Conference on Interdisciplinary Research in Engineering, Management, Pharmacy and Science at Sagar Group of Institution, Bhopal from 19/02/2015 to 22/02/2015
20.	Finite element model to study thermal variations in deep tissues of elliptical shaped human limb involving tumor.	Mamta Agrawal, K.R. Pardasani	27th International Congress of Mathematicians, (ICM-2014) COEX, Seoul, South Korea, 13- 21 Aug. 2014.
21.	FEM model to study heat flow in irregular tapered shaped human limbs involving tumors	Mamta Agrawal, K.R. Pardasani	101th Indian Science Congress held at University of Jammu, Jammu from February 3 to 7, 2014.
22.	Three dimensional finite element model to study temperature distribution in dermal regions of irregular tapered cylindrical shaped human limbs	Mamta Agrawal, K.R. Pardasani	18th Annual cum 3rd International Conference of GAMS on Mathematical, Computational and Integrative Sciences, September, 22-26, 2013 Organized by Department of Mathematics and Computer Applications, MANIT, Bhopal (M.P.), India .
23.	Three dimensional steady state temperature distribution in dermal regions of irregular tapered cylindrical shaped human limbs	Mamta Agrawal, K.R. Pardasani	17th Annual conference of GAMS and National Symposium on Computational mathematics and Information Technology December 7-9, 2012 Organized by Jaypee university of Engineering and Technology, Guna

			(M.P.)
24.	Effect of Tumor on Thermal Distribution in Elliptical Shaped Human Limbs	Mamta Agrawal Neeru Adlakha and KR Pardasani	The 7th International Conference on Computing and Information Technology, King Mongkut's University of Technology North Bangkok, Thailand 11th – 12thMay 2011
25.	Modelling and Simulation of Thermal Effect of Metastasis of Tumors in Human Limbs	Mamta Agrawal Neeru Adlakha And KR Pardasani	International Symposium on Devices Mens Intelligent Systems and Communication 2011 (ISDMISC 2011) ,Sikkim Manipal Institute of Technology, Majitar, Sikkim 12th- 14th April 2011
26.	Thermal Effect in Human Limbs involving Tumor	Mamta Agrawal Neeru Adlakha and KR Pardasani	First M.P. State IT Convention On Challenges Before Indian IT Industries In Current Scenario (CITIS) Feb. 26-27, 2011, Computer Society of India, MANIT, Bhopal
27.	Transient Temperature Distribution in Deep Tissues of Elliptical Shaped Human limbs	Mamta Agrawal Neeru Adlakha and KR Pardasani	55th congress of Indian Society of Theoretical and Applied Mechanics (ISTAM), an International meet at National Institute of Technology Hamirpur (HP),India during Dec 18-21,2010
28.	A Mathematical Model to Study Heat Flow in Deep Tissues of Elliptical Shaped Human limbs.	Mamta Agrawal Neeru Adlakha and KR Pardasani	Fifteenth Annual Conference of Gwalior Academy of Mathematical Sciences (GAMS), Mohyal Educational and Research Institute of Technology (MERIT) Dec 12-14, 2010, New Delhi.
29.	Effect of Tumor on Biothermal System in Dermal Regions of Elliptical Shaped Human Limbs.	Mamta Agrawal Neeru Adlakha and KR Pardasani	International Congress of Mathematicians (ICM) 19-27 August 2010, HICC, Hyderabad , India.
30.	Effect of Tumor on Biothermal System in Peripheral Regions of Human Limbs	Mamta Agrawal Neeru Adlakha and KR Pardasani	International Conference of women mathematicians (ICWM) 17-18 August 2010, University of Hyderabad, India.
31.	A Mathematical Model to Study Temperature Distribution in Elliptical Shaped Human Limbs Involving Non uniformly Perfused Tumor	Mamta Agrawal Neeru Adlakha and KR Pardasani	International Conference on Mathematical Computing and Management (ICMCM) 2010 , Organized by MACFAST, (Mar Athanasios College for Advanced Studies, Tiruvalla), Kerla, June 19th – June 21st , 2010.
32.	Three Dimensional Seminumerical Model to Study Temperature Distribution in Elliptical Shaped Human Limbs Involving metastasis of Tumor	Mamta Agrawal Neeru Adlakha and KR Pardasani	"Ist IFIP International Conference on Bioinformatics" during March 23rd – 29th, 2010 at SVNIT, Surat
33.	Three Dimensional Model to Study Temperature Distribution in Elliptical Shaped Human Limbs Involving Malignant Tumor	Mamta Agrawal Neeru Adlakha and KR Pardasani	International National Conference on "Emerginh Trendsd in Computer Science a Informatics" Sagar Institute of Research and Science, Bhopal March19-21, 2010
34.	Coaxial Elliptical Sector Elements to study two dimensional Heat Distribution in Dermal Regions of Human Limbs	Mamta Agrawal Neeru Adlakha and KR Pardasani	Finite Element Methods in Engineering and Science (FEMTEC 2009) Granlibakken conference center at Lake Tahoe, U.S.A.
35.	Three Dimensional Finite Element	Mamta Agrawal	National Meet for Research Scholar in

	Model to Study Thermal Disturbances due to tumor in Peripheral Regions of Elliptical Shaped Human Limbs	Neeru Adlakha and KR Pardasani	Mathematical Sciences (December 19-23, 2009) Department of Mathematics Indian Institute of Technology Roorkee-247667
36.	Finite Element Model to Study Thermal Disturbances in Two Dimensional Dermal Regions of Human Limb Involving Metastasis of Tumors	Mamta Agrawal Neeru Adlakha and KR Pardasani	14th Annual Conf of Gwalior Academy of Mathematical Sciences (GAMS), IPS Engineering college Gwalior
37.	Three dimensional seminumerical model to study heat flow in peripheral layers of elliptical shaped human limbs	Mamta Agrawal, Neeru Adlakha,and KR Pardasani	National Meet of Research Scholars in Mathematics and Statistics-2008 IIT, Kanpur 6-10 Dec. 2008
38.	Finite element model of thermograms for elliptical shaped human limbs involving tumor	Mamta Agrawal Neeru Adlakha and KR Pardasani	All India Seminar on Life Seminar on Life Cycle Analysis, Measurement & Condition Monitoring 25th-26th May, 2009, The Institution of Engineers (India), Bhopal,
39.	Seminumerical approach to study temperature distribution in peripheral layer of elliptical shape cross section of human limbs	Mamta Agrawal Neeru Adlakha and KR Pardasani	13th Annual Conf & 1st Int. Conf. of Gwalior Academy of Mathematical Sciences (GAMS) with Symp. On Mathematical Modelling in Engineering and Biosciences., Jan10-13,2008 Anand Engineering College,Agra.
40.	Coaxial Elliptical Sectoral Elements to Study Two Dimensional Heat Distribution Problems in Dermal Regions of Human Limbs.	Mamta Agrawal Neeru Adlakha and KR Pardasani	12thAnnual Conference of Gwalior Academy of Mathematical Sciences (GAMS) and All India Symposium on Computational Biology (6thApril,2007). MANIT, Bhopal

Workshop /Conferences /Symposium Attended/Training Program Attended:-

S.No.	Workshop/Conference/Symposia	Place	Year
1.	**Pydata London 2025 from 6-8June 2025**	Sancroft, Rose St, Paternoster Sq., London UK.	**6-8** June 2025
2.	OSTIV MET PANEL 2021 17-18 February 2021	Istanbul Aydın University, Istanbul, TURKEY	**17-18** February 2021
3.	Two days short term training under TEQIP-III on the subject of Mathematics, "Laplace and Fourier Transform" : Effective and Appropriate Pedagogy	Organized by RGPV, under TEQIP-II in association with Department of Applied Mathematics TIT, Bhopal	5-6th April 2019
4.	Two days National Level Workshop on big data analytics with R	Rabindranath Tagore university ,Dept.of Computer Science & Information Technology and Computer Science Engineering	from 15.03..2019 to 16.03..2019
5.	QIP Short Term Course on "Computational Methods in Fluid Mechanics and Heat Transfer"	lndian lnstitute of Technology Roorkee (IITR)Programme Centre, Roorkee '247 667	from 19.02.2018 to 23.02.2018
6.	National Workshop on Data Analytics using Spark	School of Computing Science and Engineering (SCSE), VIT	November 20-21, 2017

		Bhopal University	
7.	Participation in faculty development programme	VIT, Chennai Campus	VIT, Chennai from 12th June 2017 to 30th June 2017
8.	International Conference on Applications of Mathematics to Nonlinear Sciences (AMNS-2016)	Department of Mathematic, Kathmandu, Nepal	May 26-29, 2016
9.	Short term course on computational techniques for scientific & engineering problems	Department of Mechanical Engineering, MANIT, Bhopal	18th – 22nd May 2015
10.	International Conference on Interdisciplinary Research in Engineering, Management, Pharmacy and Science	Sagar Group of Institution, Bhopal	19/02/2015 to 22/02/2015
11.	3- day lecture program on Mathematical modelling of Biological systems (LPMMBS-2015)	Department of Mathematics University of Kashmir, srinagar	April 02-04, 2015
12	International Congress of Women Mathematicians, (ICWM-2014)	Ewha Womans University Seoul, South Korea,	12-14 Aug. 2014.
13.	27th International Congress of Mathematicians, (ICM-2014)	COEX, Seoul, South Korea,	13-21Aug. 2014.
14.	101th Indian Science Congress Organized by Indian Science congress association 14, Dr. Biresh Guha Street, Kolkata	University of Jammu, Jammu	February 3 to 7, 2014.
15.	Short term training program on Advanced Optimization Techniques for Engineers & Professionals Under TEQIP-II	Organized by Department of Mathematics and Computer Applications, MANIT , Bhopal (M.P.) India.	December 19- 21, 2013
16.	Short term training program on Mathematical Models and Methods in Engineering (Under TEQIP-II)	Organized by Department of Mathematics and Computer Applications, MANIT , Bhopal (M.P.) India.	December 16-18, 2013
17.	18th Annual cum 3rd International Conference of GAMS on Mathematical, Computational and Integrative Sciences, September, 22-26, 2013 Organized by Department of Mathematics and Computer Applications, MANIT, Bhopal, India .	Organized by Department of Mathematics and Computer Applications, MANIT , Bhopal (M.P.) India.	September, 22-26, 2013
18.	3rd IFIP International Conference on Bioinformatics under the aegis of International Federation of Information Processing (IFIP) – TC5 and Computer Society of India.	Organized by Department of Mathematics and Computer Applications, MANIT , Bhopal (M.P.) India.	September, 23-26, 2013.
19.	Short Term Course on Finite Element Modeling with Matlab under TEQIP-II (Conducted Lab on Matlab)	Jointly organized by Department of Mathematics MANIT, Bhopal and Department of Applied Mechanics, SVNIT, Surat	6th to 10th May 2013
20.	Conducted Lab on Matlab in Short Term Course on numerical Methods for Science & Engineering under TEQIP-II	Department of Mathematics MANIT, Bhopal	21st -22nd March 2013
21.	Short Term Course on numerical Methods for Science & Engineering under TEQIP-II	Department of Mathematics MANIT, Bhopal	21st -22nd March 2013
22.	Recent Advances in Water Resources management	Department of Civil Engineering MANIT, Bhopal	31st January to 1stFebruary, 2013
23.	Rainwater Harvesting–Techniques and Applications Under TEQIP II	Department of Civil Engineering MANIT, Bhopal	12-13 January

			2013
24.	Self Defence Marshal Art & Training	T.T. Nagar Stadium Bhopal	3rd January 2013 2nd February 2013 (one month)
25.	Optimization Techniques Under TEQIP II	Department of Mathematics & Computer Applications MANIT, Bhopal	10th December to11th December 2012
26.	ISTE workshop on Akash for Education	Conducted by Indian Institute of Technology Bombay under National Mission on Education through ICT (MHRD)	10th & 11th November 2012
27.	Introduction to Research Methodologies (Main Workshop)	Remote centre, MANIT, Bhopal	June 25th, 2012 to July 4th, 2012
28.	Computational Fluid Dynamics (Main Workshop)	Remote centre,MANIT, Bhopal	June 12-22, 2012
29.	National Workshop on Linear and Non-Linear Systems	Centre for Mathematical Sciences (CMS) Apaji Institute of Mathematics & Applied Computer Technology Banasthali University, Rajasthan	December 15-19, 2011
30.	UGC Sponsored-National Seminar on Recent Trends in Fluid Dynamics & Optimization Techniques and 15th Prof. P.D. Verma Memorial Lecture (NSRTFD & OT and PDVML-2011)	Department of Mathematics, University of Rajasthan, Jaipur –302004 (Raj.)	August 6-7, 2011
31.	Staff Development Program (SDP- 2011) on Mathematical Modelling and Computer Simulation Techniques	Sagar Institute of Research Technology & Sciences, Bhopal	27th June – 9th July 2011.
32.	Workshop on National language & Knowledge	Maulana Azad National Institute of Technology,Bhopal	April 15th ,2011
33.	Workshop on Soft Computing and Knowledge Mining	Department of Computer Applications and Mathematics MANIT, Bhopal	March 29th– 30th , 2011
34.	Workshop on Analytical & Numerical Solutions of Non-linear Differential Equations	Department of Mathematics, Amrita School of Engineering, Bengaluru	March 18th & 19th 2011.
35.	National Workshop on Bioinformatics	Department of Bioinformatics MANIT, Bhopal	March 11, 2011
36.	Workshop on Patent Drafting	Maulana Azad National Institute of Technology, Bhopal IPR CELL Organized	February 11-12, 2011
37.	Workshop on Data Mining and Data warehousing	Department of Maths SVNIT,Surat Gujarat	28 Dec. 2009 – Jan 1st 2010
38.	National Meet for Research Scholar in Mathematical Sciences	Department of Maths IIT Roorkee-247667	December 19-23, 2009
39	All India Seminar on "LIFE CYCLE ANALYSIS, MEASUREMENT & CONDITION MONITORING"	The Institution of Engineers (India). In MANIT, Bhopal.	May 25-26, 2009
40.	National Meet of Research Scholars in Mathematics & Statistics-2008 (NMRSMS-08)	Department of Mathematics IIT, Kanpur	6-10 Dec. 2008.
41.	EUINDIA – GRID workshop on Applications in	M.A.N.I.T., Bhopal,	5th – 9th May 2008

	Computational Biology (under TEQIP)		
42.	14th Annual Conf. of Gwalior Academy of Mathematical Sciences (GAMS) with Symp. On Computational Mathematics and its Application to Engineering, Management and Biology	IPS College of Tech. and Management, (IPS CTM),Gwalior	July 17-19, 2009
43.	National Workshop on CHEMINFORMATICS	MANIT, Bhopal	March 15th to 19th , 2008
44.	National Workshop on BIOINFORMATICS	MANIT, Bhopal	March 10th to 14th , 2008
45.	13+1 Gams & SMMB-2008 Conference held in Jaipur	Yagyavalkya Inst. Of Tech.Sitapura,	January 13, 2008
46.	13th Annual Conf & 1st Int. Conf. of Gwalior Academy of Mathematical Sciences (GAMS) with Symp. On Mathematical Modelling in Engineering & Biosciences.	Anand Engineering College, Agra.	Jan10-13,2008
47.	National Workshop on Cheminformatics	MANIT, Bhopal	May 28th–June 1st 2007
48	National Workshop on Bioinformatics	MANIT, Bhopal	21st–25th May 2007
49.	12th Annual conference of Gwalior Academy of Mathematical Sciences and All India Symposium on Computational Biology	MANIT, Bhopal	April 6-8,2007

Lecture taken in Workshop / Lab Conducted / Invited talk:-

S. No.	Title	Name of the workshop and year
1.	Conference meeting on Modelling and Machine Learning	Pydata London 2025 from 6-8June 2025 . Sancroft, Rose St, Paternoster Sq., London UK.
2.	Invited talk on "Probability, Statistics and Reliability and its Real Life Application "	Jagran School of Engineering (JSEN) under Faculty of Science And Technology, Jagran Lake city University, Bhopal 8 March 2025
3.	For contributing to the International workshop "Fairness and Trust in the Mathematical Modelling and Machine learning Techniques"	VIT Bhopal, University, 9th - 13th Oct, 2023
4.	Lecture on Heat and Mass Transfer (Speaker and Special Guest Editor"	International Conference of Mathematical Society BHU on Recent Trends in Mathematical and Computational Science (RTMCS) 3rd – 5th Feb 2023
5.	Speaker in the Mentoring & Counselling Event, India International Science Festival,	21 – 24 January 2023,India International Science Festival Organise by DBT New Delhi In MANIT Bhopal
6.	Event on National Science Day 'Indian Space Program "Future Scenario and Application"'	Organised by Jagran Lakecity University, School of Engineering and Technology 10th March 2022
7.	International Climate and Energy Forum	Istanbul Aydin University, 11th March, 2022
8	Engineering Applications of Mathematics on FEM and MATLAB	Shri G.S. Institute of Technology and Science, 1st October 2021
9.	"Interdisciplinary Mathematical Approaches in Engineering and Research" as a Guest Speaker	SAGE Group, 28th June, 2021
10.	Lecture on Research Methodology Techniques and Literature Review	RabindraNath Tagore University 19 February 2021
11.	Lecture delivered on an application of	36th Annual National Conference of the Mathematical

	biomathematics: A mathematical and computational model to study the thermal pattern in dermal regions of human limbs involving tumors"	Society NCEMS-2021 – Banaras Hindu University Varanasi 7th February 2021
12.	Guest Lecture on Mathematics Day	SAGE University 22 Dec 2020
13.	Guest Lecture	MIT Ujjain August 2020
14.	Impact of Covid-19 on higher education system	Department of Applied Science SIRTE, Bhopal, 23rd July, 2020
15.	Guest Lecture	TIT Bhopal 9th Feb 2020
16.	SIRT Bhopal 2020	SAGE University 2020
17.	Guest Lecture	SIRT Bhopal 2020
18.	Guest Lecture on Mathematical Modelling computation and simulation	Career College Bhopal on 11th January 2020
19.	"Teaching Discrete Structure: Effective & Appropriate Pedagogy" on the topic "Mathematical Logics and Its Applications in Engineering".	RGPV, Bhopal, 8th June, 2019
20.	Expert talk Topic: "Foundation of Mathematical and Propositional logic in Science and Engineering"	Short Term Training Programme on "Teaching Discrete Structure: Effective & Appropriate Pedagogy 6-7th June 2019 at Technocrats Institute of Technology Excellence Bhopal TEQIP Phase –III (RGPV Bhopal)
21.	Invited Talk: "A numerical model to study heat and water distribution in human dermal regions of upper limb".	Technocrats Institute of Technology Excellence Bhopal International Conference on Emerging Trends in Computing Technology and Management. 7-9 Feb2019.
22.	Lecture on Matlab and Lab in Dept. of Math. MANIT, Bhopal	Workshop on Mathematical Model, 24-28 June 2016
23.	Expert Lecture on Matlab in Dept. of Math. MANIT, Bhopal	Workshop on matlab for Researchers (Matlab- 13) under TEQIP-II, December 09-11, 2013
24.	Lecture on Matlab and Lab in Dept. of Math. MANIT, Bhopal	Jointly organized by Department of Mathematics MANIT, Bhopal and Department of Applied Mechanics, SVNIT, Surat, 6th May 10th May 2013.
25.	Lecture on Matlab and Lab in Department of Mathematics MANIT, Bhopal	Short Term Course on numerical Methods for Science & Engineering under TEQIP-II. 21st -22nd March 2013

Workshop/Events / Conference Organized:-

S.No.	Workshop	Place	Date
1.	For contributing to the International workshop "Fairness and Trust in the Mathematical Modelling and Machine learning Techniques"	VIT Bhopal, University,	9th - 13th Oct, 2023
2.	1st International Conference and 3rd National Conference on Recent trend in advance sciences and computing engineering as a organising secretary	VIT Bhopal, University,	17th – 19th December
3.	2nd National conference RTASCE 2020	VIT Bhopal University	18-20 Dec 2020 online
4.	International Webinar	VIT Bhopal University	23/05/2020
5.	Workshop 2 on Canvas Painting as a convener	VIT Bhopal University	03/03/2020

6.	8th International and 24th Annual Conference of Gwalior Academy of Mathematical Sciences ICGAMS-2K19 funded by DST SERB New Delhi, (As a Organizng Secretary & Convener)	VIT Bhopal University	13-15 December 2019
7.	International Pre-Conference Workshop of Gwalior Academy of Mathematical Sciences ICGAMS-2K19 funded by Mathematical Consortium on Mathematical Modelling & Simulation (As a Convener)	VIT Bhopal University	11-12 December 2019
8.	Workshop 1 on Canvas Painting as a convener	VIT Bhopal University	20/02/2019
9.	National Workshop on Fun with Mathematics as a convener	VIT Bhopal University	08/02/2019
10.	National Conference onRecent Trendsin Advanced Sciences and Computational Engineering (RTASCE 2018) as a convener	VIT Bhopal University	21-23 December 2018
11.	SPECTRA1.0 Science Fair as a co-convener	VIT Bhopal University	17th April 2018
12.	National Workshop on Research Methodology and techniques for Engineering and Management (RMTEM 2018) as a convener	VIT Bhopal University	16th August 5th-6th September 2018

Ph.D Guiding:-

1	**Venkanna Bachu :** Department of Engineering Sciences and Humanities, Kavikulguru Institute of Technology and Science, Ramtek, Nagpur, Maharashtra, 441106- India
2.	**Lakshmi Pathak :** RGPV Bhopal
3.	**Varsha Nigam :** RGPV Bhopal
4.	**Devendra Kumar :** Full time research scholar in VIT Bhopal

Life Membership of Various Societies:-

1.	Gwalior Academy of Mathematical Sciences GAMS (Member No. 143). Working Secretary of the GAMS.
2.	International Association of Engineers, IAENG. (Member No: 101262).
3.	International Association of Computer Science and Information Technology (IACSIT). (Member No. 80337076).
4.	Indian Science Congress (ISC). (Member No. L-23897).
5.	Organization for Women in Science for the Developing World (OSWD) (Member Id. 6849).
6.	Fellow membership ISROSET-FM-1045.
7.	Member of Editorial of "International Journal of Scientific Research in Mathematical and Statistical Sciences (ISSN: 2348-4519) of ISROSET group.
8.	IRED Organization Membership Info: SNM2020092039.
9.	Lepage Research Institute : Prešov Slovakia Reg. No. 50537032 http://www.lepageri.eu

Foreign Visit:

1.	**London, UK- Invitation from Pydata London 2025 from 6-8June 2025 at** Sancroft, Rose St, Paternoster Sq., London

	Support: From conference organizer free tickets for whole conference.
2.	**NEPAL, KATHMANDU- TO PRESENT PAPER (26th -29th May 2016)** **Title:** A Mathematical Model To Study Temperature Distribution In Deep Tissues Of Elliptical-Shaped Human Limbs Involving Tumor. **Travel support:** From WOS-A Project
3.	**SEOUL, SOUTH KOREA- TO PRESENT PAPER (13th - 21st August 2014).** **Title:** Finite element model to study thermal variations in deep tissues of elliptical-shaped human limb involving tumor. **DST Travel Grant :** (FILE NO. SB/ITS-S/02300/2014-15).
4.	**AUSTRIA (EUROPE) - WORK FOR INDO-AUSTRIA PROJECT (23rd June to 3rd July 2013).** **Travel support:** From Indo-Austria Project.
5.	**THAILAND, BANGKOK- TO PRESENT PAPER (11th-12th May 2011).** **Title:** Effect of Tumor on Thermal Distribution in Elliptical-Shaped Human Limbs. **DST Travel Grant:** File N0. 344/2011-12 (Travel Grant from India to Thailand May 11-12)

Certificates:-

1.	**Discrete Math and Analyzing Social Graphs** https://coursera.org/verify/9VP4N632BSYV
2.	First Steps in Linear Algebra for Machine Learning https://coursera.org/verify/3SAUK9LM4JAM
3.	Introduction to Data Science in Python https://coursera.org/verify/LEU9JTKWZHVK
4.	Python Essential by vityarthi
5.	"NEP 2020 Orientation and Programme" at Malaviya Mission Teacher Training Programme, 5th – 15th Feb, 2024
6.	"One week online FDP on Reforms in Basic Science and Humanities with References to NEP-2020" at K.D.K College of Engineering, Nagpur , 5th – 10th Feb, 2024

Language & Software:-

1.	C, C++
2.	MATLAB
3.	Mathematica
4.	Python and Machine Learning

Subject Taught:-

1.	Statistics , Probability and Reliability
2.	Calculus
3.	Differential and Difference Equation
4.	Discrete Mathematics and Graph Theory
5.	Applied Numerical Method
6.	Complex Variable and Partial Differential Equations

7.	Applied Linear Algebra
8.	Heat and Mass Transfer (Ph.D Course)
9.	Operation Research
10.	Transform Techniques and Difference Equations
11.	Business Statistics
12.	Applied Cryptography
13.	Application Differential Equations
14.	System Biology
15.	Mathematical Analysis Tools and Techniques (Ph.D Course)

Ph.D. Thesis :-

Title	" Mathematical Modelling of Temperature Distribution in Dermal Regions of Elliptical Shaped Human Limbe. "
Duration	September 2008 – September 2011
Date of Viva	12 January 2012
Supervisor Details	Dr. K.R. Pardasani (prof. MANIT Bhopal)

Master of Computer Application Project Thesis :-

Title	Sales and Purchase Management
Duration	July 1995 – July 1998

Reference:-

1. **Dr. K.R. Pardasani ,** Prof. and Former Hod, MANIT, Bhopal, Contact : +91-9425358308, Email Id : kamalrajp@rediffmail.com

2. Dr. Neeru Adlakha, Prof. Department of Applied Mathematics and Humanities ,SVNIT Surat, Gujarat , Contact : +91-8154896899, Email Id :- neeruadlakha21@gmail.com

Declaration :-

I hereby declare that the information furnished above is true to the best of my knowledge.

Place: Bhopal

Dr. Mamta Agarwal
Associate Professor
VIT Bhopal University
Bhopal, M.P